

1. Install required libraries and load the spaCy English model.

```
!pip install spacy pandas matplotlib seaborn emoji
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: emoji in /usr/local/lib/python3.12/dist-packages (2.15.0)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.21.1)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.1)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.1)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4) (2.41.4)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4) (4.14.1)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4) (0.4.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2025.11.11)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.0.1)
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (8.1.8)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (0.19.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.0.5)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.4.2->spacy) (1.17.0)
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl: 12.8/12.8 MB 62.3 MB/s eta 0:00:00
```

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

2. Load the Twitter US Airline Sentiment dataset.

```
import pandas as pd

df = pd.read_csv("Tweets.csv") # Ensure dataset is in the same folder
df.head()
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_confidence	airline	airl
0	570306133677760513	neutral	1.0000	NaN	NaN	Virgin America	
1	570301130888122368	positive	0.3486	NaN	0.0000	Virgin America	
2	570301083672813571	neutral	0.6837	NaN	NaN	Virgin America	
3	570301031407624196	negative	1.0000	Bad Flight	0.7033	Virgin America	
4	570300817074462722	negative	1.0000	Can't Tell	1.0000	Virgin America	

3. Select tweet text and sentiment columns and remove missing values.

```
df = df[['text', 'airline_sentiment']]
df.dropna(inplace=True)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14640 entries, 0 to 14639
Data columns (total 2 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   text                   14640 non-null  object
1   airline_sentiment      14640 non-null  object
dtypes: object(2)
memory usage: 228.9+ KB
```

4. Clean tweets by removing URLs, mentions, emojis, special characters, and converting text to lowercase.

```
import re
import emoji

def clean_tweet(text):
    text = text.lower()
    text = re.sub(r"http\S+|www\S+", "", text)      # URLs
    text = re.sub(r"@w+", "", text)                 # Mentions
    text = re.sub(r"#w+", "", text)                 # Hashtags (text only)
    text = emoji.replace_emoji(text, replace="")    # Emojis
    text = re.sub(r"^[^a-z\s]", "", text)           # Special characters
    text = re.sub(r"\s+", " ", text).strip()
    return text
```

5. Create a cleaned tweet corpus after preprocessing.

```
df['clean_text'] = df['text'].apply(clean_tweet)
clean_corpus = df['clean_text'].tolist()
df.head()
```

	text	airline_sentiment	clean_text
0	@VirginAmerica What @dhepburn said.	neutral	what said
1	@VirginAmerica plus you've added commercials t...	positive	plus youve added commercials to the experience...
2	@VirginAmerica I didn't today... Must mean I n...	neutral	i didnt today must mean i need to take another...
3	@VirginAmerica it's really aggressive to blast...	negative	its really aggressive to blast obnoxious enter...
4	@VirginAmerica and it's a really big bad thing...	negative	and its a really big bad thing about it

Next steps: [Generate code with df](#) [New interactive sheet](#)

6. Initialize the spaCy NLP pipeline.

```
import spacy

nlp = spacy.load("en_core_web_sm")
```

7. Create and add a custom spaCy pipeline component to detect hashtags.

```
from spacy.language import Language
from spacy.tokens import Doc

# Define custom extension
Doc.set_extension("hashtags", default=[], force=True)

@Language.component("hashtag_detector")
def hashtag_detector(doc):
    hashtags = [token.text for token in doc if token.text.startswith("#")]
    doc._.hashtags = hashtags
    return doc

# Add component to pipeline only if it's not already there
if "hashtag_detector" not in nlp.pipe_names:
    nlp.add_pipe("hashtag_detector", last=True)

nlp.pipe_names

['tok2vec',
 'tagger',
 'parser',
 'attribute_ruler',
 'lemmatizer',
 'ner',
 'hashtag_detector']
```

8. Process the cleaned tweets using the customized spaCy pipeline.

```
df['cleaned_text'] = df['text'].apply(clean_tweet)
docs = list(nlp.pipe(df["cleaned_text"]))
```

9. Extract lemmas and part-of-speech tags from processed tweets.

```
lemma_pos_data = []

for doc in docs:
    tokens = [(token.lemma_, token.pos_) for token in doc if not token.is_stop]
    lemma_pos_data.append(tokens)

lemma_pos_data[:3] # sample output

[([' ', 'SPACE'), ('say', 'VERB')],
 [(['plus', 'CCONJ'),
  ('ve', 'AUX'),
  ('add', 'VERB'),
  ('commercial', 'NOUN'),
  ('experience', 'NOUN'),
  ('tacky', 'ADV')],
 [(['not', 'PART'),
  ('today', 'NOUN'),
  ('mean', 'VERB'),
  ('need', 'VERB'),
  ('trip', 'NOUN')]]
```

10. Extract hashtags from original tweets and compute their frequencies.

```
from collections import Counter

def extract_hashtags(text):
    return re.findall(r"#\w+", text.lower())

all_hashtags = []

for tweet in df["text"]:
    all_hashtags.extend(extract_hashtags(tweet))
```

```
all_hashtags.extend(extract_hashtags(tweet))

hashtag_freq = Counter(all_hashtags)
hashtag_freq.most_common(10)
```

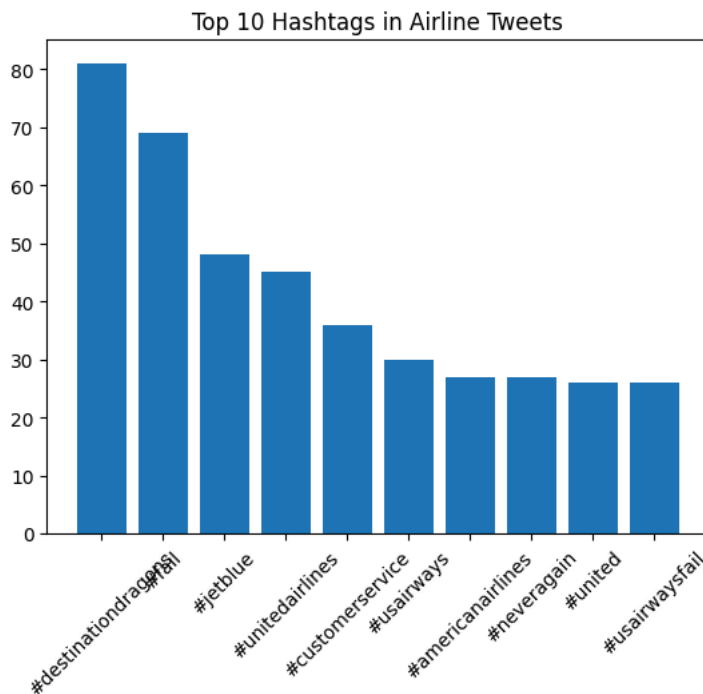
```
[('#destinationdragons', 81),
 ('#fail', 69),
 ('#jetblue', 48),
 ('#unitedairlines', 45),
 ('#customerservice', 36),
 ('#usairways', 30),
 ('#americanairlines', 27),
 ('#neveragain', 27),
 ('#united', 26),
 ('#usairwaysfail', 26)]
```

11. Visualize the most frequent hashtags.

```
import matplotlib.pyplot as plt

top_hashtags = hashtag_freq.most_common(10)
tags, counts = zip(*top_hashtags)

plt.figure()
plt.bar(tags, counts)
plt.xticks(rotation=45)
plt.title("Top 10 Hashtags in Airline Tweets")
plt.show()
```



12. Filter negative tweets and visualize their POS tag distribution.

```
negative_docs = df[df['airline_sentiment'] == 'negative']['clean_text']
negative_docs = list(nlp.pipe(negative_docs))

pos_counts = Counter()
for doc in negative_docs:
    for token in doc:
        pos_counts[token.pos_] += 1

pos_tags, pos_freq = zip(*pos_counts.most_common())

plt.figure()
plt.bar(pos_tags, pos_freq)
plt.title("POS Tag Distribution in Negative Tweets")
```

```
plt.show()
```

